Temperature-mediated Temporal Dynamics and Influence of Nectar Traits on Pollinator Visitation in Native and Non-native *Salvia*

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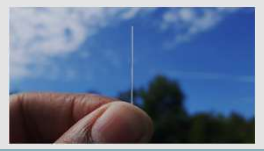
Introduction

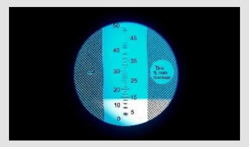
- ➢ Global pollinator decline is a major ecological concern[I].
- ➢ Urban green spaces can mitigate pollinator loss[II].
- ➢ Do floral rewards drive the pollinator visitation[III]?
- ➢ Nectar consumption by pollinators -> Depletion[VIII].
- ➢ Role of temperature in explaining pollinator visitation[IV].
- ➢ Does the visitation show a Diurnal pattern?

Methods

- Two spatial scale study – Local and Villages
- Each day a subplot was sampled.
- Sampling done thrice a day in 3 specific time blocks.
- Treatments – Bagged and Open.
- Temperature recorded by dataloggers.

Nectar Volume measurement using microcapillaries → Sugar Concentration measurement by Refractometer → Counting pollinator visitation within stipulated time





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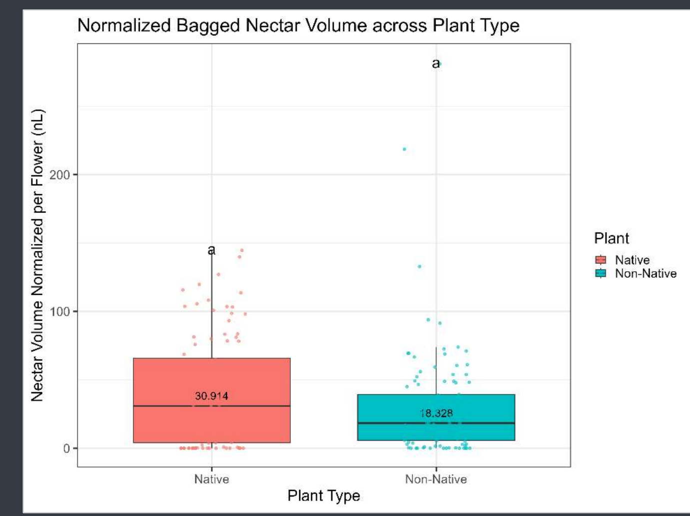
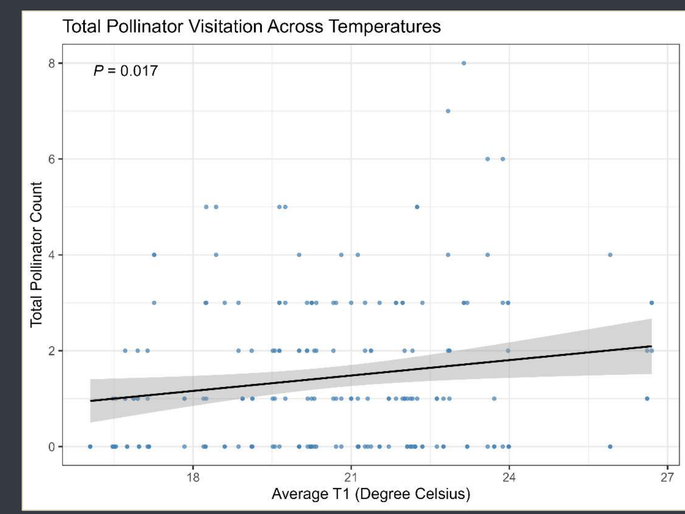
Key Results

- Nectar production was **NOT** significantly different across species.
- **TEMPERATURE** was the key factor.
- **NO** diurnal pattern observed.
- Visitation did **NOT** explain change in nectar volume.
- Depletion was **SIGNIFICANT** across species.

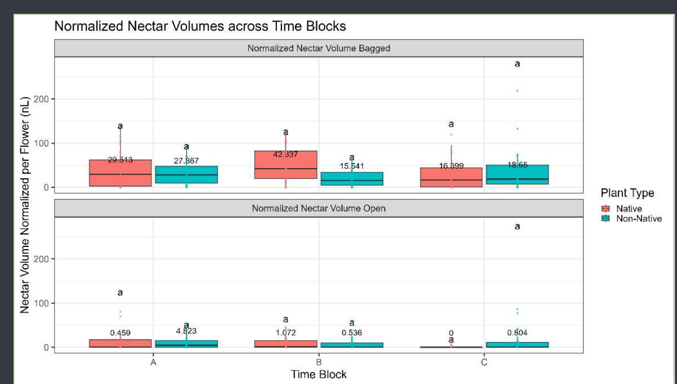
i



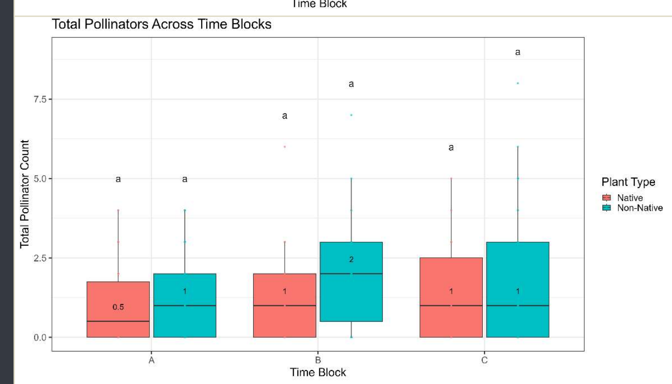
ii



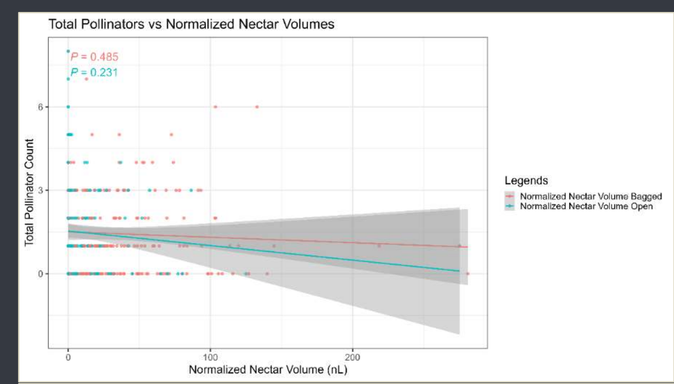
iii(a)



iii(b)



iv(a)



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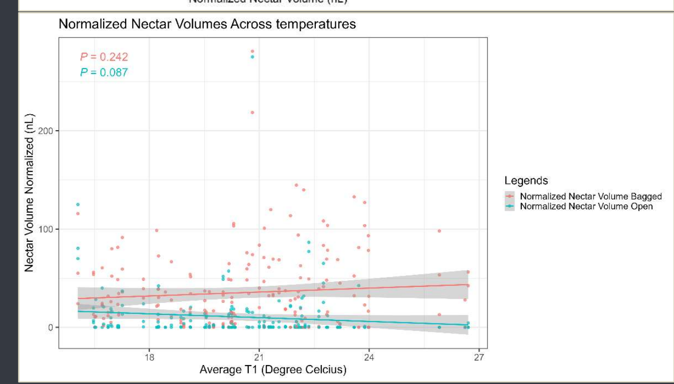
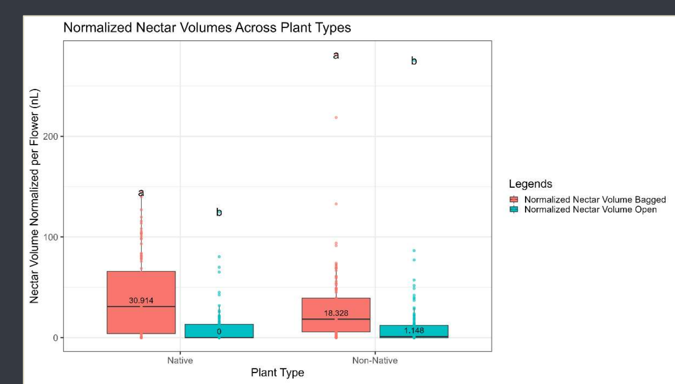


v



Discussions

- Non-native plants can integrate without significantly altering pollinator visitation[V].
- Contradicts existence of a nectar-pollinator synchrony[VI].
- High Temperature ► High Visitation
  High Temperature ► Less Standing Nectar
  Depletion more explained by evaporation that visitation.
- Pollinator visitation can be due learned familiar cues and not just floral rewards[VII].
- High depletion might indicate competition for limited resources at broader spatial scale[VIII].

Future Directions

- ➢ Diurnal dynamics needs to be seen at village scale.
- ➢ More species needs to be considered.
- ➢ Other seasons should be considered for sampling.
- ➢ Subtle methodological differences might reveal different patterns.

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